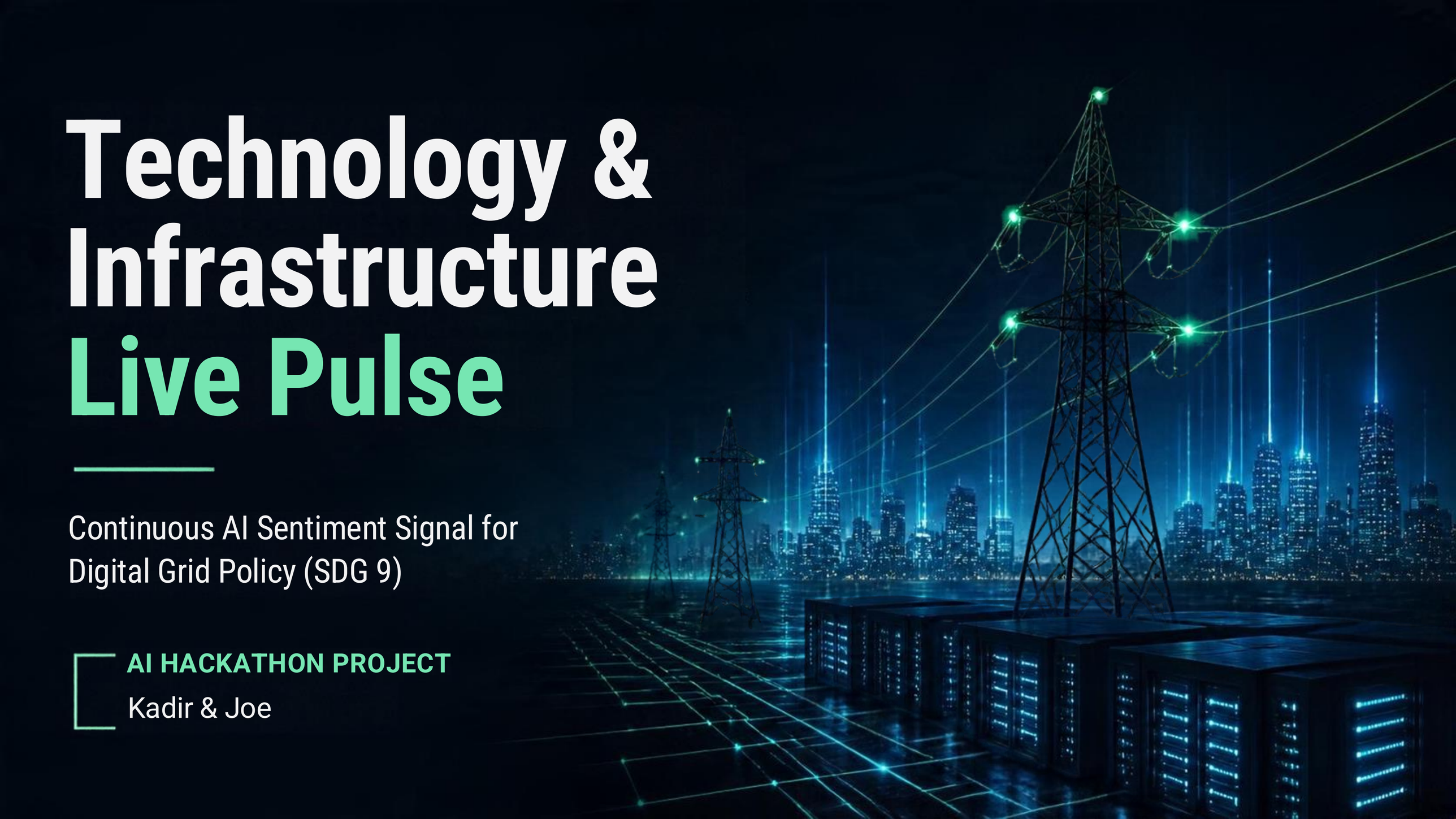
Technology & Infrastructure Live Pulse

Continuous AI Sentiment Signal for Digital Grid Policy (SDG 9)



AI HACKATHON PROJECT

Kadir & Joe

Problem & Target User

THE CHALLENGE

Rapid AI data center growth and cloud computing strain electric grids and increase household utility costs.

THE PROBLEM

Manual news tracking is slow, fragmented, and inefficient.

THE SOLUTION

Automated, real-time news radar detecting grid bottlenecks, policy announcements, and public concerns.

TARGET USER

Digital Infrastructure Policy Officer

at a local municipality



System Architecture & Pipeline

01

Data Source

Continuous Google News RSS stream fetched every 10 minutes.



02

Filtering & Deduplication

Domain keyword filters (data centers, 5G, electric grid, outages) + SHA-256 URL hashing to remove duplicates.



03

AI Sentiment Engine

Hugging Face transformer model (ProsuAI/finbert) classifying headlines as Positive, Neutral, Negative.



04

Storage & Frontend

Lightweight SQLite database powering an interactive Streamlit dashboard.



Live Dashboard in de praktijk

Deploy

Technology & Infrastructure Live Pulse

Headline sentiment monitoring for SDG 9 policy officers. Collection runs periodically. Use Refresh data to load the latest saved results.

Refresh data

Charts show headline counts grouped by collection time in UTC, not publication time or incident severity. Table filters do not change the cards or charts.

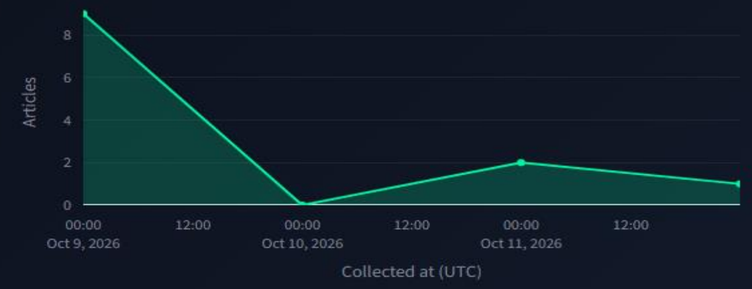
Total Monitored	Positive Headlines	Neutral Headlines	Negative Headlines
128 items	12 items	93 items	23 items

Headline Counts by Predicted Sentiment

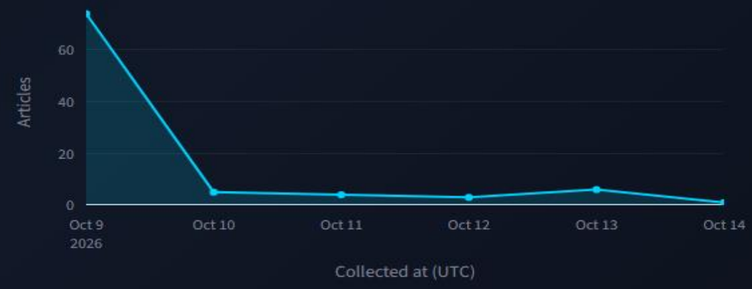
Time aggregation window:

Hourly (1h) · Every 3 Hours (3h) · Daily (24h)

Positive Headlines



Neutral Headlines



Negative Headlines



Interpretation note: Review the underlying articles when headline counts change. Sentiment predictions do not verify incidents, measure severity, or represent public opinion. This note is always displayed and is not an automatically triggered alert.

screenrec

Live Dashboard in de praktijk

🔍 Inspect Items Behind the Signal

Filter by sentiment:

positive × neutral × negative ×

unknown

Search by keyword (e.g. outage, data center, Texas):

Collected at (UTC)	Sentiment Status	News Headline	Model Confidence	Source	Original Article
14 Oct 2026, 02:22	Neutral	AI Competition Moves Into Power as 51AIpower Focuses on Individual Infrastructure Part	0.85	Google News RSS	Read News ↗
13 Oct 2026, 12:44	Neutral	'Phantom' Data Centers Are Demanding More From the Power Grid Than They Need - WS.	0.90	Google News RSS	Read News ↗
13 Oct 2026, 11:54	Negative	Trump criticizes Abbott moratorium on Texas data centers connecting to power grid - KEY	0.85	Google News RSS	Read News ↗
13 Oct 2026, 10:43	Neutral	The AI infrastructure race: As data centers outpace power grids, can storage and on-site p	0.86	Google News RSS	Read News ↗
13 Oct 2026, 08:03	Neutral	Greece Proposes 60% Renewable Electricity For Data Centers - TaiyangNews	0.81	Google News RSS	Read News ↗
13 Oct 2026, 07:13	Neutral	How Data Centers Could Spike US Electricity Bills By 2035 - Newsweek	0.81	Google News RSS	Read News ↗
13 Oct 2026, 07:13	Neutral	Can AI Data Centers Run Out of Electricity Before They Run Out of Chips? - Coinpaper	0.92	Google News RSS	Read News ↗
13 Oct 2026, 02:53	Neutral	Are Data Centers the Villains in the Battle Over Electricity? - Carnegie Endowment for Inte	0.88	Google News RSS	Read News ↗
12 Oct 2026, 15:01	Neutral	Meta to build C$13 billion Alberta data center, its first in Canada - Reuters	0.57	Google News RSS	Read News ↗
12 Oct 2026, 15:01	Negative	US power companies scramble to secure equipment as surging data center demand strai	0.94	Google News RSS	Read News ↗
12 Oct 2026, 13:31	Negative	AI data centers pose growing threat to electricity systems worldwide, UNECE says - Jurist	0.87	Google News RSS	Read News ↗
12 Oct 2026, 03:10	Positive	Kazakhstan well positioned to become regional data center hub, says expert - Qazinform	0.94	Google News RSS	Read News ↗
12 Oct 2026, 01:20	Neutral	Oregon Data Centers Now Consume 23% of State Electricity - Yahoo	0.90	Google News RSS	Read News ↗

Honest Model Benchmark

Our 50-Item Test

50	Total Headlines Evaluated from live pipeline
66.00%	Overall Accuracy 33 out of 50 correct
96.0%	Neutral Recall 24/25 very high precision on factual governance and infrastructure reports
36.8%	Negative Recall 7/19 reveals a critical neutrality blindspot regarding household utility cost issues



Ethical Reflection & Failure Analysis

CORE FAILURE

FinBERT trained on financial data misclassifies electricity price hikes as neutral economics, not public burden.

ETHICAL SAFEGUARDS

- **Human-in-the-loop:** AI signals require human verification before action.
- **Transparent links** allow users to inspect original news articles directly.
- **Strict privacy policy:** no personal or citizen data collected, only public news headlines.



Conclusion & Takeaways

AI IS ONLY AS GOOD AS ITS DATA

Confidence doesn’t equal correctness.
Training data defines what AI **can’t see.**

NEXT STEP: LOCALIZE & EXPAND

Integrate local Dutch municipal news feeds alongside international English news.

IMPACT: EMPOWER RESPONSIBLY

Enable policy officers with transparent, responsible AI decision support aligned with UN **SDG 9** goals.

